

f) $\forall x \forall y (P(x, y))$: For all students x of my class and for all computer science courses y of my school, x has taken class y .

Simple: Each student of my class has taken all the computer science courses of my school.

5) Let $W(x, y)$ mean that student x has visited website y , where the domain for x consists of all students in your school and the domain for y consists of all websites. Express each of these statements by a simple English sentence.

a) $W(\text{Sarah Smith}, \text{www.att.com})$: Sarah Smith has visited the website www.att.com .

b) $\exists x W(x, \text{www.imdb.org})$: There exists a student x from my school who has visited the website www.imdb.org .

Simple: Some student from my school has visited the website www.imdb.org .

c) $\exists y W(\text{Jose Orez}, y)$: From the set of all websites, there exists a website y which Jose Orez has visited.

Simple: Jose Orez has visited some website from the set of all websites

d) $\exists y (W(\text{Ashok Puri}, y) \wedge W(\text{Cindy Yoon}, y))$
From the set of all websites, there exists a website y s.t. Ashok Puri has visited website y and Cindy Yoon has visited website y .

Simple: Some website from the set of all websites has been visited by both Ashok Puri and Cindy Yoon.

e) $\exists y \forall z (y \neq (\text{David Belcher}) \wedge (W(\text{David Belcher}, z) \rightarrow W(y, z)))$
There exists a student y from my school such that for all websites z from the set of all websites, if y is not David Belcher and David Belcher has visited website z , then Student y has visited website z .

Simple: If David Belcher has visited all websites from the set of all websites, then there exists some other student in my school who has visited all websites. (Wrong Soln.)